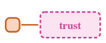
CONCEPT	DIMENSION	CODING
Self-referential data Which data refers to itself, and what sits inside one node?	DATA	**List** HtDP DCIC A list whose rest is again a list. 
		Tree Software Construction ×2 A tree whose children are trees. 
		Nested doll HtDP A pictorial object holding a smaller copy. 
	NODE CONTENT	**Concrete data** DCIC Software Construction Values of the input itself. [1, 4, 3]
		Call with arguments SICP LeetCode ×3 The call as written, concrete args. fib 3
		State tuple LeetCode ×3 Only the changing parameters. (i, k)
		Partial expression SICP HtDP LeetCode Half-evaluated term, as in a trace. (* 6 (f 5))
Base case Is the stopping point shown, and how?	MARKING	**Marked** LeetCode ×4 Labelled or highlighted as the base case.
		Reached, not marked LYAH The last call is drawn but not singled out. f(1) → f(0)
		Dead-end leaf LeetCode A failed branch ends the recursion. 

CONCEPT	DIMENSION	CODING		
Decomposition How are the smaller instances laid out, and how many calls are drawn?	STRUCTURE	**Nested containment** Calls drawn inside their caller.	*Box-in-box* HtDP 	
			Brackets LYAH LeetCode ×2 f(a(b(c)))	
		Tree (node-link) Calls linked to their subcalls.	*Call tree* SICP LeetCode ×3 	
			State tree LeetCode ×2 	
			Choice tree LeetCode 	
			Result tree LeetCode ×6 	
		Linear chain List of nodes and links.	*Box-and-pointer* HtDP LeetCode 	
			Pointer rewiring LeetCode ×8 	
		Stack LeetCode ×2 Frames piled up by depth. 		
		Table HtDP LeetCode Args, intermediates and results. 		
		Substitution trace SICP HtDP ×3 LeetCode Expression rewritten until a value. 		
		Indentation staircase DCIC LeetCode Depth shown by indentation. 		
		Partition LeetCode Input split into boxed parts. 		
		Instance mapping LeetCode ×3 Size n aligned to size n–1. 		
		Metaphor / pictorial LYAH HtDP LeetCode Dolls, mirrors, spirals. 		
	CHOICES	**Choice label** LeetCode ×2 The decision named on the edge. 		
		Prune / fail + legend LeetCode ×3 Dead branches, keyed in a legend. 		
	ELISION (CALLS)	**Full expansion** SICP LeetCode ×2 Small input, every call drawn. 		
		Patterned elision CLRS Rule implied by a few levels. 		
		Unpatterned elision LeetCode No inference can be made. 		
		Deferred call Subcall left unexpanded.	*Trace term* HtDP f(rest…)	
			Done box LeetCode ×2 done	
		Trusted subcall LeetCode ×5 The subcall's result is used without drawing inside it (leap of faith). 		

CONCEPT	DIMENSION	CODING
Return & combine How do results come back, and how many returns are drawn?	FLOW	**Upward only** HtDP Results merging toward the root. 
		Round trip SICP LeetCode ×7 Down to the base case, then back up. 
		Rewrite in place SICP HtDP ×2 Time runs down the lines. f(3) ⇒ 3·f(2) ⇒ 3·2·f(1) ⇒ 6
	MARKS	**Return value** LeetCode ×3 Value written on the way back. 
		Combine step SICP HtDP LeetCode ×2 How sub-results merge is drawn. 
		Traversal path SICP Evaluation order traced. 
		Call-step label LeetCode ×2 Each frame or step numbered. call 1 call 2
	ELISION (RETURNS)	**All returns drawn** SICP ×2 HtDP ×3 LeetCode ×7 Every call's result is shown coming back.
		Some returns drawn LeetCode Only some results, e.g. the final one.
		Returns not drawn LeetCode ×4 Calls only; for a recursive process the way back is missing. 

LYAH Learn You a Haskell
SICP Structure and Interpretation of Computer Programs
HtDP How to Design Programs
DCIC A Data-Centric Introduction to Computing
CLRS Introduction to Algorithms
Software Construction MIT Software Construction course notes
LeetCode LeetCode solutions, vote ≥ 5
×n number of examples

Figure X: A design space for recursion diagrams in three tables. Each recursion concept lists the dimensions along which it is drawn.